HW6

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1

$$Y = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \end{pmatrix}$$

X is n by num_treatments. Which 'treatment' is each response from?

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}$$

Z is n by num_blocks. Which 'block' is each response from?

$$Z = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$e = \begin{pmatrix} e_{11} \\ e_{12} \\ e_{13} \\ e_{21} \\ e_{22} \\ e_{23} \end{pmatrix}$$

2

$$G = \begin{pmatrix} \sigma_b^2 & 0 \\ 0 & \sigma_b^2 \end{pmatrix}$$

$$R = \begin{pmatrix} \sigma^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2 \end{pmatrix}$$

$$ZGZ' = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sigma_b^2 & 0 \\ 0 & \sigma_b^2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \sigma_b^2 & 0 \\ \sigma_b^2 & 0 \\ \sigma_b^2 & 0 \\ 0 & \sigma_b^2 \\ 0 & \sigma_b^2 \\ 0 & \sigma_b^2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix} =$$

$$\begin{pmatrix} \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \end{pmatrix}$$

Now add R:

$$V(Y) = ZGZ' + R = \begin{pmatrix} \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2 \end{pmatrix} + \begin{pmatrix} \sigma^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2 \end{pmatrix} =$$

$$\begin{pmatrix} \sigma_b^2+\sigma^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2+\sigma^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2+\sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2+\sigma^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2+\sigma^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2+\sigma^2 \end{pmatrix}$$

3

$K = \begin{pmatrix} 1/2 & 0 & 0 & 1/2 & 0 & 0 \end{pmatrix}$

$KY = y_{11}/2 + y_{21}/2 = \bar{y}_{.1}$

$V(KY) = KV(Y)K' = K(ZGZ' + R)K'$

$$\begin{pmatrix} 1/2 & 0 & 0 & 1/2 & 0 & 0 \end{pmatrix} \begin{pmatrix} \sigma_b^2+\sigma^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2+\sigma^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2+\sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2+\sigma^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2+\sigma^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2+\sigma^2 \end{pmatrix} \begin{pmatrix} 1/2 \\ 0 \\ 0 \\ 1/2 \\ 0 \\ 0 \end{pmatrix} =$$

$$\begin{pmatrix} (\sigma_b^2+\sigma^2)/2 & \sigma_b^2/2 & \sigma_b^2/2 & (\sigma_b^2+\sigma^2)/2 & \sigma_b^2/2 & \sigma_b^2/2 \end{pmatrix} \begin{pmatrix} 1/2 \\ 0 \\ 0 \\ 1/2 \\ 0 \\ 0 \end{pmatrix} = (\sigma_b^2+\sigma^2)/4 + (\sigma_b^2+\sigma^2)/4 =$$

$(\sigma_b^2 + \sigma^2)/2$

$K_1 = \begin{pmatrix} 1/2 & -1/2 & 0 & 1/2 & -1/2 & 0 \end{pmatrix}$

$V(K_1Y) = K_1V(Y)K_1' = K_1(ZGZ' + R)K_1'$

$$\begin{pmatrix} 1/2 & -1/2 & 0 & 1/2 & -1/2 & 0 \end{pmatrix} \begin{pmatrix} \sigma_b^2+\sigma^2 & \sigma_b^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2+\sigma^2 & \sigma_b^2 & 0 & 0 & 0 \\ \sigma_b^2 & \sigma_b^2 & \sigma_b^2+\sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_b^2+\sigma^2 & \sigma_b^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2+\sigma^2 & \sigma_b^2 \\ 0 & 0 & 0 & \sigma_b^2 & \sigma_b^2 & \sigma_b^2+\sigma^2 \end{pmatrix} \begin{pmatrix} 1/2 \\ -1/2 \\ 0 \\ 1/2 \\ -1/2 \\ 0 \end{pmatrix} =$$

$$\begin{pmatrix} \sigma^2/2 & -\sigma^2/2 & -\sigma^2/2 & \sigma^2/2 & -\sigma_b^2/2 & -\sigma^2/2 \end{pmatrix} \begin{pmatrix} 1/2 \\ -1/2 \\ 0 \\ 1/2 \\ -1/2 \\ 0 \end{pmatrix} = \sigma^2/4 + \sigma^2/4 + \sigma^2/4 + \sigma^2/4 = \sigma^2$$

4

$$Z = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What happens to G?

$$G = \begin{pmatrix} \sigma_b^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_b^2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_s^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_s^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_s^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_s^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \sigma_s^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_s^2 \end{pmatrix}$$

5-8

At this point I'm combining 5-8 all into one because I can just do all 10 repetitions at once, then output the overall table with the final proportion of 'captured' events.

Block 1: Treatment 1, Treatment 2, Treatment 3

Block 2: Treatment 1, Treatment 2, Treatment 3

Block 3: Treatment 1, Treatment 2, Treatment 3

Block 4: Treatment 1, Treatment 2, Treatment 3

$y_{ijk} = tmt_i + block_j + error_{ijk}$

```
suppressPackageStartupMessages(library(lme4))
suppressPackageStartupMessages(library(emmeans))

set.seed(1337)
treat_means <- c(25, 35, 30)
sig2_e <- 10
sig2_b <- 8
n_treat <- 3
n_block <- 4
n_samp <- 4
u <- rnorm(4, 0, sqrt(sig2_b))

gen_data <- function() {
    treatment <- factor(paste0("Trt_", 1:3))
    block <- factor(paste0("Blk_", 1:4))
    reps <- 1:4

    # Create the 48-row design matrix
    design_matrix <- expand.grid(Rep = reps, Treatment = treatment, Block = block)

    y <- unlist(lapply(1:n_treat, \(i) lapply(1:n_block, \(j) lapply(1:n_samp, \(k) treat_mea
    dat <- as.data.frame(cbind(y, design_matrix$Block, design_matrix$Treatment))
    colnames(dat) <- c("y", "Block", "Treatment")
    dat$Block <- as.factor(dat$Block)
    dat$Treatment <- as.factor(dat$Treatment)
    dat
}

one_rep <- function(i) {
    dt <- gen_data()
    mod <- lmer(y ~ Treatment + (1|Block), data = dt)
    trt_means <- emmeans(mod, ~ Treatment)
    ci_table <- data.frame(summary(trt_means, infer = c(TRUE, FALSE), df = 35, level = 0.95))
    ci_table$true_val <- treat_means
    ci_table$captured <- ci_table$true_val >= ci_table$lower.CL & ci_table$true_val <= ci_tab
    ci_table
}

dt_list <- lapply(1:10, one_rep)
```

```r
final_table <- do.call(rbind, dt_list)

mean_coverage <- mean(final_table$captured)
```

Full table of results:

```r
final_table
```

```
   Treatment   emmean       SE df lower.CL upper.CL true_val captured
1          1 30.68615 2.273983 35 26.06972 35.30258       25    FALSE
2          2 31.86884 2.273983 35 27.25241 36.48527       35     TRUE
3          3 30.77349 2.273983 35 26.15706 35.38992       30     TRUE
4          1 29.58938 2.807836 35 23.88916 35.28959       25     TRUE
5          2 31.88139 2.807836 35 26.18118 37.58160       35     TRUE
6          3 30.06762 2.807836 35 24.36741 35.76783       30     TRUE
7          1 30.55312 2.558992 35 25.35809 35.74815       25    FALSE
8          2 30.02791 2.558992 35 24.83288 35.22294       35     TRUE
9          3 29.72732 2.558992 35 24.53229 34.92235       30     TRUE
10         1 28.57752 2.715165 35 23.06544 34.08960       25     TRUE
11         2 31.78613 2.715165 35 26.27405 37.29821       35     TRUE
12         3 29.84998 2.715165 35 24.33790 35.36206       30     TRUE
13         1 28.86163 2.422637 35 23.94341 33.77984       25     TRUE
14         2 30.25786 2.422637 35 25.33964 35.17607       35     TRUE
15         3 29.20493 2.422637 35 24.28672 34.12315       30     TRUE
16         1 28.79680 2.344131 35 24.03796 33.55563       25     TRUE
17         2 30.24496 2.344131 35 25.48612 35.00380       35     TRUE
18         3 30.29360 2.344131 35 25.53476 35.05244       30     TRUE
19         1 28.62568 2.558842 35 23.43096 33.82041       25     TRUE
20         2 31.94054 2.558842 35 26.74582 37.13527       35     TRUE
21         3 29.72932 2.558842 35 24.53459 34.92404       30     TRUE
22         1 28.61971 2.646706 35 23.24661 33.99280       25     TRUE
23         2 30.62097 2.646706 35 25.24787 35.99407       35     TRUE
24         3 30.13111 2.646706 35 24.75801 35.50420       30     TRUE
25         1 28.40383 2.721645 35 22.87859 33.92906       25     TRUE
26         2 29.57684 2.721645 35 24.05161 35.10207       35     TRUE
27         3 30.65373 2.721645 35 25.12850 36.17896       30     TRUE
28         1 28.16690 2.596419 35 22.89589 33.43791       25     TRUE
29         2 32.16277 2.596419 35 26.89176 37.43378       35     TRUE
30         3 29.20553 2.596419 35 23.93452 34.47654       30     TRUE
```

Coverage of the intervals:

```
mean_coverage
```

```
[1] 0.9333333
```

That's pretty dang close to the expected 95% coverage we'd expect (it's actually the closest we could possibly get, with either 28/30 or 29/30 covering).